Fourier Transform

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Fourier Integral Theorem

A periodic function $f(x)$ defined in a finite interval $(-L, L)$ can be expressed in Fourier series. By extending this concept, non-periodic functions defined in $-\infty < x < \infty$ (for all x) can be expressed as a Fourier integral, since in practice periodic functions are fairly rare.

Fourier Integral Theorem

A periodic function $f(x)$ defined in a finite interval $(-L, L)$ can be expressed in Fourier series. By extending this concept, non-periodic functions defined in $-\infty < x < \infty$ (for all x) can be expressed as a Fourier integral, since in practice periodic functions are fairly rare.

Theorem (Fourier Integral Theorem)

A function $f(x)$, which is piecewise continuous in every finite interval and is absolutely integrable on the x-axis, can be represented by a Fourier integral

$$f(x) = \int_0^\infty [A(w)\cos wx + B(w)\sin wx]dw \tag{1}$$

which is valid at all points of continuity. At a point of discontinuity x_0 , the Fourier integral $= \frac{1}{2}[f(x_0 - 0) + f(x_0 + 0)]$ i.e., average of the left and right hand limits of $f(x)$ at x_0, where

$$A(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\cos(wt)\ dt \quad \text{and} \quad B(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\sin(wt)\ dt$$

Note

Substituting the values of $A(w)$ and $B(w)$ in (1) yields

$$f(x) = \int_0^\infty \left[\left(\frac{1}{\pi} \int_{-\infty}^{\infty} f(t)\cos(wt)dt \right) \cos wx + \left(\frac{1}{\pi} \int_{-\infty}^{\infty} f(t)\sin(wt)dt \right) \sin wx \right] dw$$

$$= \frac{1}{\pi} \int_0^\infty \left[\int_{-\infty}^{\infty} f(t)[\cos(wt)\cos wx + \sin(wt)\sin wx]dt \right] dw$$

$$= \frac{1}{\pi} \int_0^\infty \left[\int_{-\infty}^{\infty} f(t)\cos w(t-x)dt \right] dw$$

Thus

$$f(x) = \frac{1}{\pi} \int_0^\infty \left[\int_{-\infty}^{\infty} f(t)\cos w(t-x)dt \right] dw$$

This is another form of Fourier integral.

Fourier Cosine Integral

- When $f(x)$ is an even function, then $B(w) = 0$ and

$$A(w) = \frac{2}{\pi} \int_0^\infty f(t) \cos(wt)\ dt$$

Then the Fourier integral (1) reduces to the Fourier **cosine integral**

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$$f(x) = \int_0^\infty B(w) \sin wx\ dw,$$

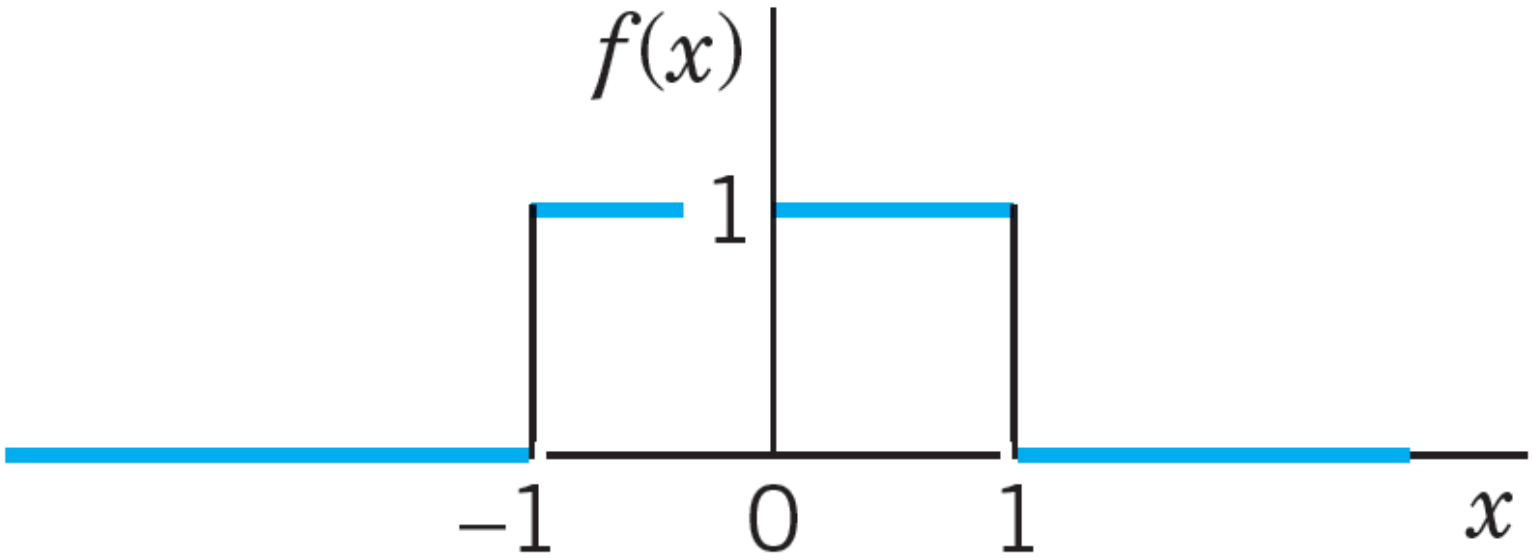
where

$$B(w) = \frac{2}{\pi} \int_0^\infty f(t) \sin(wt)\ dt$$

Example

Find the Fourier integral representation of the function

$$f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases}$$



Solution

$$A(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\cos(wt)\ dt = \frac{1}{\pi}\int_{-1}^{1} \cos(wt)\ dt = \frac{\sin(wt)}{\pi w}\bigg|_{-1}^{1} = \frac{2\sin w}{\pi w}$$

$$B(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\sin(wt)\ dt = \frac{1}{\pi}\int_{-1}^{1} \sin(wt)\ dt = \frac{\cos(wt)}{\pi w}\bigg|_{-1}^{1} = 0$$

Therefore the Fourier integral representation of $f(x)$ is

$$f(x) = \int_0^{\infty} [A(w)\cos wx + B(w)\sin wx]dw = \frac{2}{\pi}\int_0^{\infty} \frac{\sin w\cos wx}{w}\ dw \qquad (2)$$

The average of the left- and right-hand limits of $f(x)$ at $x = 1$ is equal to $\frac{1+0}{2}$, that is, $\frac{1}{2}$. Furthermore, from Fourier integral theorem

$$\int_0^{\infty} \frac{\sin w\cos wx}{w}\ dw = \begin{cases} \frac{\pi}{2} & \text{if } 0 \le x < 1, \\ \frac{\pi}{4} & \text{if } x = 1, \\ 0 & \text{if } x > 1. \end{cases}$$

We mention that this integral is called **Dirichlet's discontinuous factor**.

Remark

If we put $x = 0$ in (2), then

$$f(0) = \frac{2}{\pi} \int_0^\infty \frac{\sin w}{w}\, dw$$

$$\implies 1 = \frac{2}{\pi} \int_0^\infty \frac{\sin w}{w}\, dw$$

$$\implies \int_0^\infty \frac{\sin w}{w}\, dw = \frac{\pi}{2}$$

We see that this integral is the limit of the so-called **sine integral** ,i.e., $\int_0^t \frac{\sin w}{w}\, dw$.

Example

Find the Fourier integral representation of

$$f(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$$

and find the value of the resulting integral when

(a) $x < 0$,

(b) $x = 0$, and

(c) $x > 0$.

Solution

$$A(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\cos(wt)\ dt = \frac{1}{\pi}\int_{0}^{\infty} e^{-t}\cos(wt)\ dt = \frac{1}{\pi(1+w^2)}$$

$$B(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\sin(wt)\ dt = \frac{1}{\pi}\int_{0}^{\infty} e^{-t}\sin(wt)\ dt = \frac{w}{\pi(1+w^2)}$$

Therefore, the Fourier integral representation of $f(x)$ is

$$f(x) = \int_0^{\infty}[A(w)\cos(wx)+B(w)\sin(wx)]dw = \frac{1}{\pi}\int_0^{\infty}\frac{\cos(wx)+w\sin(wx)}{1+w^2}\ dw.$$

Applying the Fourier integral theorem to this integral, we find that

$$\int_0^{\infty}\frac{\cos(wx)+w\sin(wx)}{1+w^2}\ dw = \begin{cases} 0 & \text{if } x<0 \\ \frac{\pi}{2} & \text{if } x=0 \\ \pi e^{-x} & \text{if } x>0 \end{cases}$$

When $x = 0$, this last result is seen to reduce to the familiar definite integral

$$\int_0^{\infty}\frac{dw}{1+w^2}dw = \frac{\pi}{2}.$$

Example

Find the Fourier sine integral for $f(x) = e^{-\beta x},\ \beta > 0$ and hence show that

$$\frac{\pi}{2} e^{-\beta x} = \int_0^\infty \frac{w \sin wx}{w^2 + \beta^2}\, dw.$$

Solution

$$B(w) = \frac{2}{\pi}\int_0^\infty f(t)\sin(wt)\,dt = \frac{2}{\pi}\int_0^\infty e^{-\beta t}\sin(wt)\,dt$$
$$= \frac{2}{\pi}\left[\frac{e^{-\beta t}}{w^2+\beta^2}[-\beta\sin(wt) - w\cos(wt)]\right]_0^\infty$$
$$= \frac{2w}{\pi(w^2+\beta^2)}$$

Thus, the Fourier sine integral of $f(x) = e^{-\beta x}$ is

$$f(x) = \int_0^\infty B(w)\sin(wx)\,dw = \frac{2}{\pi}\int_0^\infty \frac{w\sin(wx)}{w^2+\beta^2}\,dw.$$

Putting the value of $f(x)$ and simplifying, we get

$$\int_0^\infty \frac{w\sin(wx)}{w^2+\beta^2}\,dw = \frac{\pi}{2}e^{-\beta x}.$$

Example

Express

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq \pi \\ 0 & \text{if } x > \pi \end{cases}$$

as a Fourier sine integral and hence evaluate

$$\int_0^\infty \frac{1 - \cos(\pi w)}{w} \sin(xw)\, dw.$$

Solution

$$B(w) = \frac{2}{\pi}\int_0^{\infty} f(t)\sin(wt)\ dt = \frac{2}{\pi}\int_0^{\pi} \sin(wt)\ dt$$
$$= \frac{2}{\pi}\left[\frac{-\cos(wt)}{w}\right]_0^{\pi} = \frac{2}{\pi}\left[\frac{1-\cos(\pi w)}{w}\right]$$

Thus, the Fourier sine integral of $f(x)$ is

$$f(x) = \int_0^{\infty} B(w)\sin(wx)\ dw = \frac{2}{\pi}\int_0^{\infty} \frac{1-\cos(\pi w)}{w}\sin wx\ dw.$$

Therefore

$$\int_0^{\infty} \frac{1-\cos(\pi w)}{w}\sin wx\ dw = \frac{\pi}{2}f(x).$$

At $x = \pi$, which is a point of discontinuity of $f(x)$, the value of the above integral is $\frac{\pi}{2}\left(\frac{1+0}{2}\right) = \frac{\pi}{4}$. Thus

$$\int_0^{\infty} \frac{1-\cos(\pi w)}{w}\sin wx\ dw = \begin{cases} \frac{\pi}{2} & \text{if } 0 \le x < \pi \\ \frac{\pi}{4} & \text{if } x = \pi \\ 0 & \text{if } x > \pi \end{cases}$$

Example

Find the Fourier cosine and Fourier sine integrals of $f(x) = e^{-kx}$, where $x > 0$ and $k > 0$.

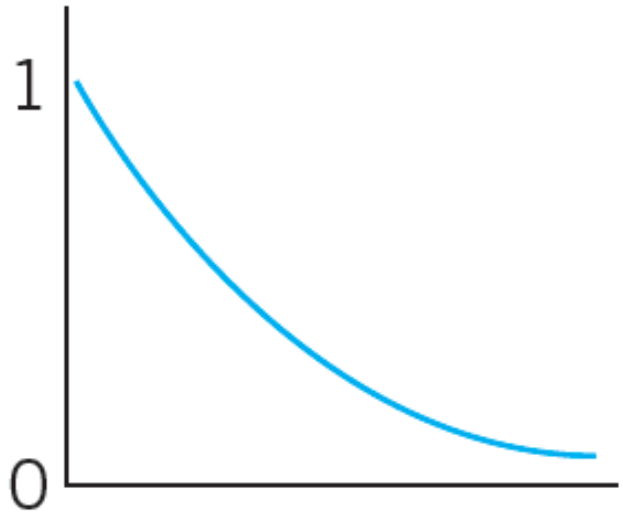


Figure 1: The function $f(x) = e^{-kx}$.

Solution

$$A(w) = \frac{2}{\pi}\int_0^\infty f(t)\cos wt\ dt = \frac{2}{\pi}\int_0^\infty e^{-kt}\cos wt\ dt$$
$$= \frac{2}{\pi}\left[-\frac{e^{-kw}}{k^2+w^2}(k\cos wt - w\sin wt)\right]_0^\infty = \frac{2}{\pi}\frac{k}{k^2+w^2}$$

we thus obtain the Fourier cosine integral representation

$$f(x) = e^{-kx} = \frac{2}{\pi}\int_0^\infty \frac{k}{k^2+w^2}\cos(wx)\ dw$$

From this representation, we see that

$$\int_0^\infty \frac{\cos(wx)}{k^2+w^2}\ dw = \frac{\pi}{2k}e^{-kx}$$

Solution cont. . .

$$B(w) = \frac{2}{\pi}\int_0^\infty f(t)\sin wt\ dt = \frac{2}{\pi}\int_0^\infty e^{-kt}\sin wt\ dt$$
$$= \frac{2}{\pi}\left[-\frac{e^{-kw}}{k^2+w^2}(w\cos wt + k\sin wt)\right]_0^\infty = \frac{2}{\pi}\frac{w}{k^2+w^2}$$

we thus obtain the Fourier sine integral representation

$$f(x) = e^{-kx} = \frac{2}{\pi}\int_0^\infty \frac{k}{k^2+w^2}\cos(wx)\ dw$$

From this representation, we see that

$$\int_0^\infty \frac{u\sin(wx)}{k^2+w^2}\ dw = \frac{\pi}{2}e^{-kx}$$

Exercise 1:

Find the Fourier integral representation of $f(x) = e^{-|x|}$.

Ans: $f(x) = \frac{2}{\pi} \int_0^\infty \frac{\cos(wx)}{1+w^2}\, dw$

Exercise 2:

Find the Fourier integral representation of the function $f(x) = \begin{cases} 0 & \text{if } |x| > a \\ \frac{bx}{a} & \text{if } |x| \leq a \end{cases}$.

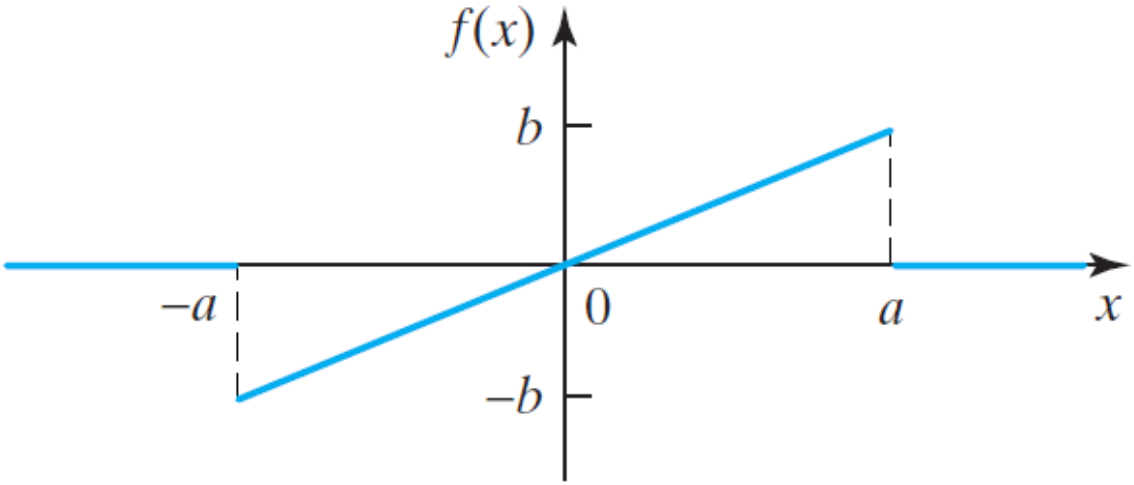


Figure 2: The truncated straight line function

Ans: $A(w) = 0,\ B(w) = \frac{2b}{\pi a w^2}[\sin(aw) - aw\cos(aw)]$ **and**

$$f(x) = \frac{2}{\pi a}\int_0^\infty \frac{[\sin(aw) - aw\cos(aw)]\sin(wx)}{w^2}\, dw$$

Exercise 3:

Find the Fourier integral representation of the function

$$f(x) = \begin{cases} \frac{\pi}{2}\cos x & \text{if } |x| < \frac{\pi}{2} \\ 0 & \text{if } |x| > \frac{\pi}{2} \end{cases}.$$

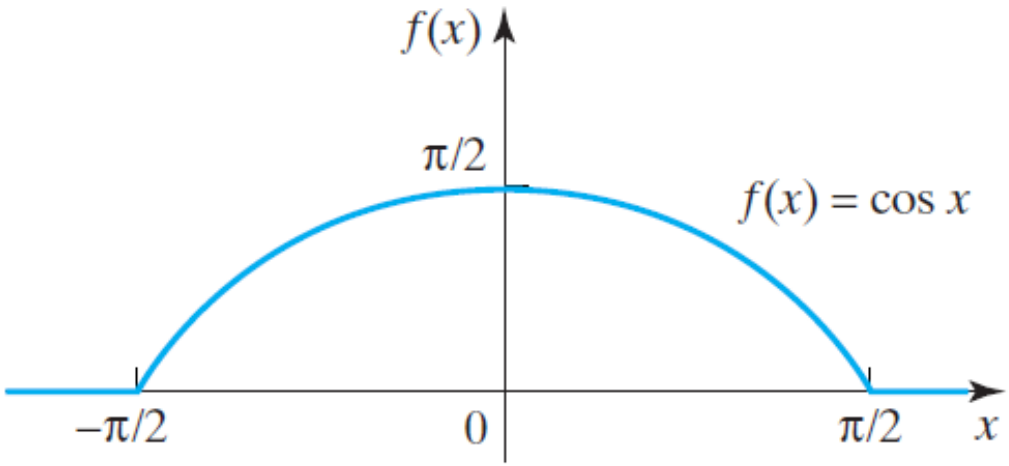


Figure 3: The truncated cosine function.

Ans: $f(x) = \int_0^\infty \frac{\cos\frac{\pi w}{2}\cos wx}{1-w^2}\,dw.$

Exercise 4:

Find the Fourier integral representation of the function

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ \cos x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}.$$

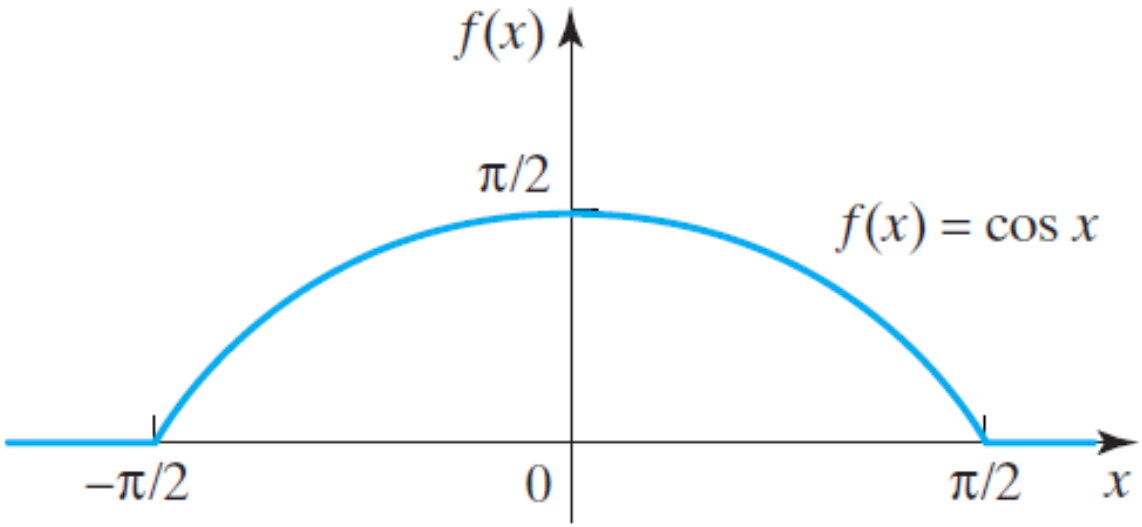


Figure 4: The asymmetric truncated cosine function.

Ans: $f(x) = \frac{1}{\pi} \int_0^\infty \frac{w[\sin(wx) - \sin(\pi + x)w]}{w^2 - 1}\, dw.$

Exercise 5:

Find the Fourier cosine integral representation of the following functions

(a) $f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$ **Ans:** $f(x) = \frac{2}{\pi} \int_0^\infty \frac{\sin w \cos wx}{w}\, dw$

(b) $f(x) = \begin{cases} \sin x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$ **Ans:** $f(x) = \frac{2}{\pi} \int_0^\infty \frac{\cos(\pi w) + 1}{1 - w^2} \cos(wx)\, dw$

(c) $f(x) = \begin{cases} x^2 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$

Exercise 6:

Find the Fourier sine integral representation of the following functions

(a) $f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$ **Ans:** $f(x) = \frac{2}{\pi} \int_0^\infty \frac{1 - \cos w}{w} \sin wx \; dw$

(b) $f(x) = \begin{cases} e^x & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$

Ans: $f(x) = \frac{2}{\pi} \int_0^\infty \frac{w - e(w \cos w - \sin w)}{1 + w^2} \sin wx \; dw$

Fourier Cosine Transform

The Fourier cosine transform concerns even functions. We obtain it from the Fourier cosine integral

$$f(x) = \int_0^\infty A(w)\cos wx \; dw, \qquad \text{where} \quad A(w) = \frac{2}{\pi}\int_0^\infty f(t)\cos(wt) \; dt$$

Namely, we set $A(w) = \sqrt{\frac{2}{\pi}}\hat{f}_c(w)$, where c suggests "cosine." Then, writing $t = x$ in the formula for $A(w)$, we have

$$\hat{f}_c(w) = \mathscr{F}_c(f) = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\cos(wx) \; dx \tag{3}$$

and

$$f(x) = \sqrt{\frac{2}{\pi}}\int_0^\infty \hat{f}_c(w)\cos(wx) \; dw \tag{4}$$

Formula (3) gives from $f(x)$ a new function $\hat{f}_c(w)$, called the **Fourier cosine transform** of $f(x)$. Formula (4) gives us back $f(x)$ from $\hat{f}_c(w)$ and we therefore call $f(x)$ the **inverse Fourier cosine transform** of $\hat{f}_c(w)$ and denote it by $\mathscr{F}_c{}^{-1}(\hat{f}_c(w))$.

Fourier Sine Transform

The Fourier sine transform concerns odd functions. We obtain it from the Fourier sine integral

$$f(x) = \int_0^\infty B(w) \sin wx \; dw, \qquad \text{where} \quad B(w) = \frac{2}{\pi} \int_0^\infty f(t) \sin(wt) \; dt$$

Namely, we set $B(w) = \sqrt{\frac{2}{\pi}} \hat{f}_s(w)$, where s suggests "sine." Then, writing $t = x$ in the formula for $A(w)$, we have

$$\hat{f}_s(w) = \mathscr{F}_s(f) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin(wx) \; dx \tag{5}$$

and

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_s(w) \sin(wx) \; dw \tag{6}$$

Formula (5) gives from $f(x)$ a new function $\hat{f}_s(w)$, called the **Fourier sine transform** of $f(x)$. Formula (6) gives us back $f(x)$ from $\hat{f}_s(w)$ and we therefore call $f(x)$ the **inverse Fourier sine transform** of $\hat{f}_s(w)$ and denote it by $\mathscr{F}_s{}^{-1}(\hat{f}_s(w))$.

Example

Find the Fourier cosine and Fourier sine transforms of the function

$$f(x) = \begin{cases} k & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

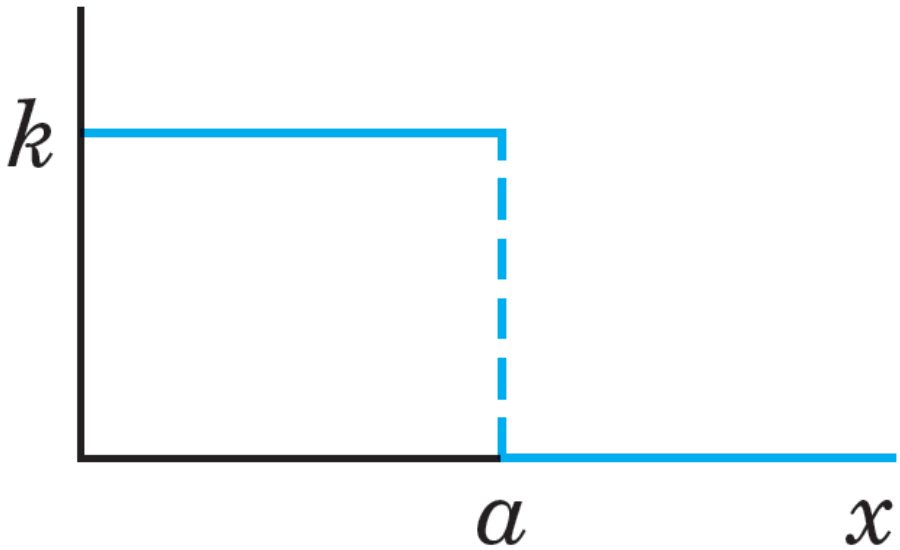


Figure 5: Plot of the function $f(x)$.

Solution

From definitions of cosine and sine Fourier transforms, we have

$$\mathscr{F}_c(f) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos(wx)\ dx = \sqrt{\frac{2}{\pi}} k \int_0^{a} \cos(wx)\ dx$$

$$= \sqrt{\frac{2}{\pi}} k \left[\frac{\sin wx}{w}\right]_0^a$$

$$= \sqrt{\frac{2}{\pi}} k \left(\frac{\sin aw}{w}\right)$$

Solution

From definitions of cosine and sine Fourier transforms, we have

$$\mathscr{F}_c(f) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos(wx)\ dx = \sqrt{\frac{2}{\pi}} k \int_0^a \cos(wx)\ dx$$
$$= \sqrt{\frac{2}{\pi}} k \left[\frac{\sin wx}{w}\right]_0^a$$
$$= \sqrt{\frac{2}{\pi}} k \left(\frac{\sin aw}{w}\right)$$

$$\mathscr{F}_s(f) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin(wx)\ dx = \sqrt{\frac{2}{\pi}} k \int_0^\infty \sin(wx)\ dx$$
$$= \sqrt{\frac{2}{\pi}} k \left[\frac{-\cos wx}{w}\right]_0^a$$
$$= \sqrt{\frac{2}{\pi}} k \left(\frac{1-\cos aw}{w}\right)$$

Example

Find $\mathscr{F}_c(e^{-x})$.

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Solution

By definition of cosine Fourier transform, we get

$$\begin{aligned}\mathscr{F}_c(e^{-x}) &= \sqrt{\frac{2}{\pi}}\int_0^\infty e^{-x}\cos(wx)\,dx \\ &= \sqrt{\frac{2}{\pi}}\left[\frac{e^{-x}}{1+w^2}(-\cos wx + w\sin wx)\right]_0^\infty \\ &= \sqrt{\frac{2}{\pi}}\left(\frac{1}{1+w^2}\right)\end{aligned}$$

Example

Find $\mathscr{F}_s(e^{-ax})$.

Example

Find $\mathscr{F}_s(e^{-ax})$.

Solution

By definition of sine Fourier transform, we get

$$\begin{aligned}\mathscr{F}_s(e^{-ax}) &= \sqrt{\frac{2}{\pi}}\int_0^\infty e^{-ax}\sin(wx)\,dx \\ &= \sqrt{\frac{2}{\pi}}\left[\frac{e^{-x}}{a^2+w^2}(-a\sin wx - w\cos wx)\right]_0^\infty \\ &= \sqrt{\frac{2}{\pi}}\left(\frac{w}{a^2+w^2}\right)\end{aligned}$$

Existance of Fourier cosine and sine transforms

Existance of Fourier cosine and sine transforms

If $f(x)$ is absolutely integrable on the positive x-axis and piecewise continuous on every finite interval, then the Fourier cosine and sine transforms of f exist.

Linearity property of Fourier cosine and sine transforms

Linearity property of Fourier cosine and sine transforms

If f and g have Fourier cosine and sine transforms, so does $af + bg$ for any constants a and b, and

- $\mathscr{F}_c(af + bg) = a\mathscr{F}_c(f) + b\mathscr{F}_c(g)$.
- $\mathscr{F}_s(af + bg) = a\mathscr{F}_s(f) + b\mathscr{F}_s(g)$.

Fourier Cosine and Sine Transforms of Derivatives

Fourier Cosine and Sine Transforms of Derivatives

Let $f(x)$ be a continuous and absolutely integrable on the x-axis, let $f'(x)$ be piecewise continuous on every finite interval, and let $f(x) \to 0$ as $x \to \infty$. Then

- $\mathscr{F}_c(f'(x)) = w\mathscr{F}_s(f(x)) - \sqrt{\frac{2}{\pi}} f(0)$.
- $\mathscr{F}_s(f'(x)) = -w\mathscr{F}_c(f(x))$.

Fourier Cosine and Sine Transforms of Derivatives

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Let $f(x)$ be a continuous and absolutely integrable on the x-axis, let $f'(x)$ be piecewise continuous on every finite interval, and let $f(x) \to 0$ as $x \to \infty$. Then

- $\mathscr{F}_c(f'(x)) = w\mathscr{F}_s(f(x)) - \sqrt{\frac{2}{\pi}} f(0).$
- $\mathscr{F}_s(f'(x)) = -w\mathscr{F}_c(f(x)).$

Fourier Cosine and Sine Transforms of second Derivatives

If f', f'' satisfy the respective assumptions for f, f' in above result. Then

- $\mathscr{F}_c(f''(x)) = -w^2\mathscr{F}_c(f(x)) - \sqrt{\frac{2}{\pi}} f'(0).$
- $\mathscr{F}_s(f''(x)) = -w^2\mathscr{F}_s(f(x)) + \sqrt{\frac{2}{\pi}} wf(0).$

Example

Find the Fourier cosine transform $\mathscr{F}_c(e^{-ax})$ of $f(x) = e^{-ax}$, where $a > 0$.

Example

Find the Fourier cosine transform $\mathscr{F}_c(e^{-ax})$ of $f(x) = e^{-ax}$, where $a > 0$.

Solution

By Differentiation, $\dfrac{d^2}{dx^2}e^{-ax} = a^2 e^{-ax}$; thus

$$a^2 f(x) = f''(x)$$

Therefore, the linearity property yields

$$a^2 \mathscr{F}_c(f) = \mathscr{F}_c(f'')$$

$$\implies a^2 \mathscr{F}_c(f) = -w^2 \mathscr{F}_c(f) - \sqrt{\frac{2}{\pi}} f'(0)$$

$$\implies (a^2 + w^2)\mathscr{F}_c(f) = \sqrt{\frac{2}{\pi}} a$$

$$\implies \mathscr{F}_c(f) = \sqrt{\frac{2}{\pi}} \left(\frac{a}{a^2 + w^2} \right)$$

Example

Find the Fourier sine transform of $f(x) = \dfrac{e^{-ax}}{x}$ and hence find $\mathscr{F}_s\left(\dfrac{e^{-ax} - e^{-bx}}{x}\right)$.

Solution

$$\mathscr{F}_s(f) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin(wx)\ dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{e^{-ax}}{x} \sin(wx)\ dx$$

$$\implies \frac{d}{dw}\mathscr{F}_s(f) = \sqrt{\frac{2}{\pi}} \frac{d}{dw}\left(\int_0^\infty \frac{e^{-ax}}{x} \sin(wx)\ dx \right)$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-ax} \cos(wx)\ dx$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-ax}}{a^2 + w^2}(-a\cos wx + w \sin wx) \right]_0^\infty$$

$$\implies \frac{d}{dw}\mathscr{F}_s(f) = \sqrt{\frac{2}{\pi}} \left(\frac{a}{a^2 + w^2} \right)$$

Integrating, we get

$$\mathscr{F}_s(f) = \sqrt{\frac{2}{\pi}} \tan^{-1}\left(\frac{w}{a}\right) + C$$

Solution cont. . .

Thus, we have

$$\mathscr{F}_s(f) = \sqrt{\frac{2}{\pi}} \tan^{-1}\left(\frac{w}{a}\right) + C$$

$$\Longrightarrow \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{e^{-ax}}{x} \sin(wx)\, dx = \sqrt{\frac{2}{\pi}} \tan^{-1}\left(\frac{w}{a}\right) + C$$

Putting $w = 0$ yields $C = 0$. Therefore

$$\mathscr{F}_s\left(\frac{e^{-ax}}{x}\right) = \sqrt{\frac{2}{\pi}} \tan^{-1}\left(\frac{w}{a}\right)$$

Now

$$\mathscr{F}_s\left(\frac{e^{-ax} - e^{-bx}}{x}\right) = \mathscr{F}_s\left(\frac{e^{-ax}}{x}\right) - \mathscr{F}_s\left(\frac{e^{-bx}}{x}\right)$$

$$= \sqrt{\frac{2}{\pi}} \left[\tan^{-1}\left(\frac{w}{a}\right) - \tan^{-1}\left(\frac{w}{b}\right)\right] = \sqrt{\frac{2}{\pi}} \tan^{-1}\left[\frac{(b-a)w}{ab + w^2}\right]$$

Example

Find the inverse Fourier cosine transform of $\dfrac{\sin aw}{w}$.

Example

Find the inverse Fourier cosine transform of $\dfrac{\sin aw}{w}$.

Solution

Given that $\hat{f}_c(w) = \frac{\sin aw}{w}$ and we are required to find $f(x)$. Therefore

$$
\begin{aligned}
f(x) &= \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_c(w) \cos wx \; dw \\
&= \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{\sin aw}{w} \cos wx \; dw \\
&= \frac{1}{\sqrt{2\pi}} \int_0^\infty \frac{\sin(a+x)w + \sin(a-x)w}{w} dw \\
&= \frac{1}{\sqrt{2\pi}} \left[\int_0^\infty \frac{\sin(a+x)w}{w} dw + \int_0^\infty \frac{\sin(a-x)w}{w} dw \right]
\end{aligned}
$$

Solution cont...

Using the result $\int_0^\infty \frac{\sin ax}{x}\, dx = \begin{cases} \frac{\pi}{2} & \text{if } a > 0 \\ -\frac{\pi}{2} & \text{if } a < 0 \end{cases}$, we obtain

$$f(x) = \frac{1}{\sqrt{2\pi}} \begin{cases} \frac{\pi}{2} + \frac{\pi}{2} & \text{if } a + x > 0 \text{ and } a - x > 0 \\ 0 & \text{if } a < 0 \end{cases}$$

$$= \begin{cases} \sqrt{\frac{\pi}{2}} & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

Exercise 7:

Find the Fourier cosine transform $\hat{f}_c(w)$ of $f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ -1 & \text{if } 1 < x < 2 \\ 0 & \text{if } x > 2 \end{cases}$.

Ans: $\hat{f}_c(w) = \sqrt{\frac{2}{\pi}} \left(\frac{2 \sin w - \sin 2w}{w} \right)$

Exercise 8:

Find $\hat{f}_c(w)$ for $f(x) = \begin{cases} x & \text{if } 0 < x < 2 \\ 0 & \text{if } x > 2 \end{cases}$.

Ans: $\hat{f}_c(w) = \sqrt{\frac{2}{\pi}} \left(\frac{\cos 2w + 2w \sin 2w - 1}{w^2} \right)$

Exercise 9:

Find $\hat{f}_c(w)$ for $f(x) = \begin{cases} x^2 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$.

Ans: $\hat{f}_c(w) = \sqrt{\frac{2}{\pi}} \left[\frac{(w^2 - 2) \sin w + 2w \cos w}{w^3} \right]$

Exercise 10:

Find the Fourier cosine transform of the function

$$f(x) = \begin{cases} \cos x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}.$$

Ans: $\hat{f}_c(w) = \dfrac{1}{\sqrt{2\pi}} \left[\dfrac{\sin a(w+1)}{w+1} + \dfrac{\sin a(w-1)}{w-1} \right]$

Exercise 11:

Does the Fourier cosine transform of $x^{-1} \sin x$ $(0 < x < \infty)$ exist? Of $x^{-1} \cos x$ $(0 < x < \infty)$? Give reasons.

Exercise 12:

Find $\hat{f}_s(w)$ for $f(x) = \begin{cases} x^2 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$.

Ans: $\hat{f}_s(w) = \sqrt{\dfrac{2}{\pi}} \left[\dfrac{(2 - w^2)\cos w + 2w \sin w - 2}{w^3} \right]$

Exercise 13:

Find the inverse Fourier transform of

$$\mathscr{F}_s(f(x)) = \frac{e^{-aw}}{w}, \ a > 0$$

Ans: $f(x) = \sqrt{\frac{2}{\pi}} \tan^{-1}\left(\frac{x}{a}\right)$

Exercise 14:

Find the function $f(x)$ such that

$$\int_0^\infty f(x) \sin wx \; dx = \begin{cases} 1 & \text{if } 0 \leq w < 1 \\ 2 & \text{if } 1 \leq w < 2 \\ 0 & \text{if } w \geq 2 \end{cases} .$$

Solution

Since $\sin wx$ is present in the given integral, we have

$$\mathscr{F}_s(f) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin wx \, dx = \sqrt{\frac{2}{\pi}} \begin{cases} 1 & \text{if } 0 \le w < 1 \\ 2 & \text{if } 1 \le w < 2 \\ 0 & \text{if } w \ge 2 \end{cases} = \begin{cases} \sqrt{\frac{2}{\pi}} & \text{if } 0 \le w < 1 \\ 2\sqrt{\frac{2}{\pi}} & \text{if } 1 \le w < 2 \\ 0 & \text{if } w \ge 2 \end{cases}$$

Therefore, $f(x)$ is inverse Fourier sine transform of $\mathscr{F}_s(f)$. Thus

$$\begin{aligned} f(x) &= \sqrt{\frac{2}{\pi}} \int_0^\infty \mathscr{F}_s(f) \sin wx \; dw \\ &= \frac{2}{\pi} \left[\int_0^1 \sin wx \; dw + \int_1^2 2 \sin wx \; dw \right] \\ &= \frac{2}{\pi} \left[-\frac{\cos wx}{x} \Big|_0^1 - 2 \frac{\cos wx}{x} \Big|_1^2 \right] \\ f(x) &= \frac{2}{\pi} \left(\frac{1 + \cos x - 2 \cos 2x}{x} \right) \end{aligned}$$

Exercise 15:

Find the function $f(x)$, satisfying the integral equation

$$\int_0^\infty f(x) \sin wx \; dx = \frac{w}{w^2 + k^2}$$

Fourier transform

Fourier transform

The **Fourier transform** of the function $f(x)$ is denoted by $\hat{f}(w)$ or $\mathscr{F}(f)$ and is defined as

$$\hat{f}(w) = \mathscr{F}(f) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} \, dx.$$

and

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(w) e^{iwx} \, dw.$$

is called **inverse Fourier transform**.

Existence of the Fourier Transform

If $f(x)$ is absolutely integrable on the x-axis and piecewise continuous on every finite interval, then the Fourier transform of $f(x)$ given by

$$\mathscr{F}(f) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} \, dx$$

exists.

Example

Find the Fourier transform of

$$f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{otherwise} \end{cases}$$

and hence evaluate $\int_0^\infty \frac{\sin x}{x}\, dx$.

Solution

By definition, we get

$$\begin{aligned}
\mathscr{F}(f) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{-iwx}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-iwx}}{-iw} \right]_{-1}^{1} \\
&= \frac{1}{\sqrt{2\pi}} \frac{1}{w} \left(\frac{e^{-iw} - e^{iw}}{i} \right) \\
&= \sqrt{\frac{2}{\pi}} \frac{\sin w}{w}
\end{aligned}$$

Solution cont. . .

Now, by inverse Fourier transform, we have

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathscr{F}(f) e^{iwx}\, dw = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin w}{w} e^{iwx}\, dw$$

Putting $x = 0$ yields

$$f(0) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin w}{w}\, dw = \frac{2}{\pi} \int_{0}^{\infty} \frac{\sin w}{w}\, dw$$

$$\implies 1 = \frac{2}{\pi} \int_{0}^{\infty} \frac{\sin w}{w}\, dw$$

$$\implies \frac{\pi}{2} = \int_{0}^{\infty} \frac{\sin w}{w}\, dw$$

Thus

$$\boxed{\int_{0}^{\infty} \frac{\sin x}{x}\, dx = \frac{\pi}{2}}$$

Example

Find the Fourier transform $\mathscr{F}(f)$ of $f(x)$ if

$$f(x) = \begin{cases} e^{-ax} & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases},$$

here $a > 0$.

Solution

By definition, we get

$$\begin{aligned}
\mathscr{F}(f) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-ax} e^{-iwx}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-(a+iw)x}}{-(a+iw)} \right]_0^{\infty} \\
&= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{a+iw} \right)
\end{aligned}$$

Exercise 16:

The function $f(x)$ is defined by

$$f(x) = e^{-a|x|},$$

where a is a positive constant. Find the Fourier transform of $f(x)$.

Ans: $\mathscr{F}(e^{-a|x|}) = \sqrt{\frac{2}{\pi}}\ \frac{a}{a^2 + w^2}$.

Solution

$$\mathscr{F}(e^{-a|x|}) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} e^{ax} e^{-iwx} dx + \frac{1}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-ax} e^{-iwx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{(a-iw)x}}{a-iw} \Big|_{-\infty}^{0} - \frac{e^{-(a+iw)x}}{a+iw} \Big|_{0}^{\infty} \right]$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{1}{a-iw} + \frac{1}{a+iw} \right] = \sqrt{\frac{2}{\pi}} \frac{a}{a^2+w^2}$$

Thus

$$\mathscr{F}^{-1}\left(\frac{a}{a^2+w^2} \right) = \sqrt{\frac{\pi}{2}} e^{-a|x|}$$

Exercise 17:

The triangle function $\Lambda_n(x)$ is defined as

$$\Lambda_n = \begin{cases} \frac{1}{n^2}(n+x), & -n < x < 0 \\ \frac{1}{n^2}(n-x), & 0 < x < n \\ 0 & \text{otherwise} \end{cases}$$

where n is a positive constant.

(a) Sketch the graph of $\Lambda_n(x)$.

(b) Show that the Fourier transform of $\Lambda_n(x)$ is

$$\frac{1}{\sqrt{2\pi}} \operatorname{sinc}^2\left(\frac{1}{2}wn\right),$$

where $\operatorname{sinc} x = \frac{\sin x}{x}$.

Exercise 18:

The function $f(x)$ is defined in terms of the positive constant a, by

$$f(x) = \begin{cases} 1 - \frac{|x|}{a} & \text{if } |x| \leq a \\ 0 & \text{if } |x| > a \end{cases}$$

Find the Fourier transform of $f(x)$.

Ans: $\mathscr{F}(f(x)) = \sqrt{\frac{2}{\pi}} \frac{1}{aw^2}[1 - \cos(aw)]$

Exercise 19:

The function $f(x)$ is defined by

$$f(x) = \begin{cases} mx & \text{if } |x| \leq \frac{1}{m} \\ 0 & \text{if } |x| > \frac{1}{m} \end{cases},$$

where m is a positive constant. Find the Fourier transform of $f(x)$.

Ans: $\hat{f}(w) = \frac{1}{w}\sqrt{\frac{2}{\pi}}\left[\cos\left(\frac{w}{m}\right) - \text{sinc}\left(\frac{w}{m}\right)\right]$

Exercise 20:

The function f is defined by

$$f(x) = \frac{1}{x}, \quad x \neq 0.$$

Determine the Fourier transform of $f(x)$, assuming without proof any standard results about $\int_0^\infty \frac{\sin ax}{x}\, dx$.

Ans: $\mathscr{F}\left(\frac{1}{x}\right) = -i\sqrt{\frac{\pi}{2}}\ sign(w)$, where signum function $sign(x)$ is defined as

$$sign(x) = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

Exercise 21:

Find the Fourier transform of an arbitrary function $f(x)$ if

(i) $f(x)$ is even.

(ii) $f(x)$ is odd.

Give the answers as a simplified integral form.

Ans: If $f(x)$ is even then $\hat{f}(w) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(t) \cos wx \; dx$ and if $f(x)$ is odd then $\hat{f}(w) = -i\sqrt{\frac{2}{\pi}} \int_0^\infty f(t) \sin wx \; dx$

Example

Find the Fourier transform of $f(x) = e^{-ax^2}$.

Hint: $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.

Using this, find the Fourier transform of e^{-x^2}.

Solution

$$\begin{aligned}
\mathscr{F}(e^{-ax^2}) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-ax^2}e^{-iwx}\,dx \\
&= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(ax^2+iwx)}\,dx \\
&= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{\left[-(ax^2+iwx)+\left(\frac{iw}{2\sqrt{a}}\right)^2-\left(\frac{iw}{2\sqrt{a}}\right)^2\right]}\,dx \\
&= \frac{e^{-\frac{w^2}{4a}}}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\left(\sqrt{a}x+\frac{iw}{2\sqrt{a}}\right)^2}\,dx
\end{aligned}$$

Putting $\sqrt{a}x + \frac{iw}{2\sqrt{a}} = t \implies dx = \frac{dt}{\sqrt{a}}$, we get

$$\mathscr{F}(e^{-ax^2}) = \frac{e^{-\frac{w^2}{4a}}}{\sqrt{2\pi a}}\int_{-\infty}^{\infty} e^{-t^2}\,dt = \frac{e^{-\frac{w^2}{4a}}}{\sqrt{2\pi a}}\sqrt{\pi}$$

$$\mathscr{F}(e^{-ax^2}) = \frac{1}{\sqrt{2a}}e^{-\frac{w^2}{4a}}$$

Solution cont. . .

We have

$$\mathscr{F}(e^{-ax^2}) = \frac{1}{\sqrt{2a}} e^{-\frac{w^2}{4a}}$$

Putting $a = 1$ yields

$$\mathscr{F}(e^{-x^2}) = \frac{1}{\sqrt{2}} e^{-\frac{w^2}{4}}$$

Definition(Self reciprocal)

If $\mathscr{F}(f(x)) = f(w)$,then the function $f(x)$ is called self reciprocal.

Definition(Self reciprocal)

If $\mathscr{F}(f(x)) = f(w)$,then the function $f(x)$ is called self reciprocal.

Exercise 24:

Prove that Fourier transform of $e^{-\frac{x^2}{2}}$ is self reciprocal.

Exercise 25:

Show that $\int_{-\infty}^{\infty} e^{-x^2} \cos x \, dx = \sqrt{\pi} e^{-\frac{1}{4}}$. You can use the fact that $\mathscr{F}(e^{-ax^2}) = \frac{1}{\sqrt{2a}} e^{-\frac{w^2}{4a}}$.

Properties of Fourier Transform

Linearity of the Fourier Transform

The Fourier transform is a linear operation; that is, for any functions $f(x)$ and $g(x)$ whose Fourier transforms exist and any constants a and b, the Fourier transform of $af(x) + bg(x)$ exists, and

$$\mathscr{F}(af(x) + bg(x)) = a\mathscr{F}(f(x)) + b\mathscr{F}(g(x)).$$

Properties of Fourier Transform cont...

Fourier Transform of the Derivative of $f(x)$

* Let $f(x)$ be continuous on the x-axis and $f(x) \to 0$ as $|x| \to \infty$. Furthermore, let $f'(x)$ be absolutely integrable on the x-axis. Then

$$\mathscr{F}(f'(x)) = iw\mathscr{F}(f(x)).$$

Properties of Fourier Transform cont...

Fourier Transform of the Derivative of $f(x)$

- Let $f(x)$ be continuous on the x-axis and $f(x) \to 0$ as $|x| \to \infty$. Furthermore, let $f'(x)$ be absolutely integrable on the x-axis. Then

$$\mathscr{F}(f'(x)) = iw\mathscr{F}(f(x)).$$

- Let $f(x), f'(x), \ldots f^{n-1}(x)$ all are continuous on the x-axis and they all tend to zero as $|x| \to \infty$. Furthermore, let $f^{(j)}(x)$ be absolutely integrable on the x-axis for each $j = 0, 1, 2, \ldots, n$, then

$$\mathscr{F}(f^{(n)}(x)) = (iw)^n\mathscr{F}(f(x)).$$

Properties of Fourier Transform cont...

Shifting property

If $\mathscr{F}(f(x)) = F(w)$ and a is any real number then

$$\mathscr{F}(f(x-a)) = e^{-iwa}F(w)$$

Properties of Fourier Transform cont...

Shifting property

If $\mathscr{F}(f(x)) = F(w)$ and a is any real number then

$$\mathscr{F}(f(x-a)) = e^{-iwa}F(w)$$

Frequency Shifting Property

If $\mathscr{F}(f(x)) = F(w)$ and w_0 is any real number then

$$\mathscr{F}\left\{f(x)e^{iw_0x}\right\} = F\left(w - w_0\right)$$

Properties of Fourier Transform cont...

Change of scale property

If $\mathscr{F}(f(x)) = F(w)$ and a is any real number then

$$\mathscr{F}\{f(ax)\} = \frac{1}{|a|} F\left(\frac{w}{a}\right)$$

Properties of Fourier Transform cont...

Change of scale property

If $\mathscr{F}(f(x)) = F(w)$ and a is any real number then

$$\mathscr{F}\{f(ax)\} = \frac{1}{|a|} F\left(\frac{w}{a}\right)$$

Integration Property

If $\mathscr{F}(f(x)) = F(w)$ then

$$\mathscr{F}\left\{\int_{-\infty}^{x} f(\tau)d\tau\right\} = \frac{1}{iw} F(w)$$

Properties of Fourier Transform cont...

Multiplication by x property

- If $\mathscr{F}(f(x)) = F(w)$ then

$$\mathscr{F}\{xf(x)\} = iF'(w)$$

Properties of Fourier Transform cont...

Multiplication by x property

- If $\mathscr{F}(f(x)) = F(w)$ then

$$\mathscr{F}\{xf(x)\} = iF'(w)$$

- If $\mathscr{F}(f(x)) = F(w)$ then

$$\mathscr{F}\{x^n f(x)\} = i^n F^{(n)}(w)$$

Example

Find the Fourier transform of xe^{-x^2}.

Example

Find the Fourier transform of xe^{-x^2}.

Solution

Since we know that

$$\mathscr{F}\{xf(x)\} = iF'(w)$$

Therefore

$$\mathscr{F}(e^{-x^2}) = i\frac{d}{dw}\left(\frac{1}{\sqrt{2}}e^{-\frac{w^2}{4}}\right)$$
$$= -\frac{iw}{2\sqrt{2}}e^{-\frac{w^2}{4}}$$

Example

Find the Fourier transform of $f(x) = e^{-ax^2}$.

Hint: $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.

Solution

We have already found the Fourier transform of $f(x) = e^{-ax^2}$ but here we'll find the Fourier transform of $f(x) = e^{-ax^2}$ using the properties of the Fourier transform. Given that

$$\begin{aligned} & f(x) = e^{-ax^2} \\ \implies & f'(x) = -2axe^{-ax^2} = -2axf(x) \\ \implies & f'(x) + 2axf(x) = \end{aligned}$$

Taking Fourier transforms of both sides, we get

$$\begin{aligned} & \mathscr{F}(f'(x)) + 2a\mathscr{F}(xf(x)) = 0 \\ \implies & iw\hat{f}(w) + 2ai\frac{d\hat{f}(w)}{dw} = 0 \qquad \text{where } \hat{f}(w) = \mathscr{F}(f(x)) \\ \implies & 2a\frac{d\hat{f}(w)}{dw} + w\hat{f}(w) = 0, \end{aligned}$$

which is a first order linear differential equation (variable-separable). Solving this, we get

$$\hat{f}(w) = Ae^{-\frac{w^2}{4a}}$$

Solution cont. . .

where A is an arbitrary constant. Note that

$$A = \hat{f}(0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ax^2} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(\sqrt{a}x)^2} dx$$

Putting $\sqrt{a}x = t \implies dx = \frac{dt}{\sqrt{a}}$. Thus

$$A = \frac{1}{\sqrt{2\pi a}} \int_{-\infty}^{\infty} e^{-t^2} dt = \frac{1}{\sqrt{2\pi a}} \sqrt{\pi} = \frac{1}{\sqrt{2a}}$$

Therefore

$$\hat{f}(w) = \mathscr{F}(e^{-ax^2}) = \frac{1}{\sqrt{2a}} e^{-\frac{w^2}{4a}}$$

Example

Find the solution of the differential equation

$$\frac{dy}{dt} - 2y = \begin{cases} 0 & \text{if} \quad t < 0 \\ e^{-2t} & \text{if} \quad t \geq 0 \end{cases}, \quad -\infty < t < \infty,$$

using Fourier transforms.

Solution

Given Differential equation

$$\frac{dy}{dt} - 2y = \begin{cases} 0 & \text{if} \quad t < 0 \\ e^{-2t} & \text{if} \quad t \geq 0 \end{cases}$$

Taking Fourier transform of both sides, we get

$$\mathscr{F}\left(\frac{dy}{dt}\right) - 2\mathscr{F}(y) = \mathscr{F}\left(\begin{cases} 0 & \text{if} \quad t < 0 \\ e^{-2t} & \text{if} \quad t \geq 0 \end{cases}\right)$$

$$\implies iw\hat{y}(w) - 2\hat{y}(w) = \frac{1}{\sqrt{2\pi}}\frac{1}{2+iw}$$

$$\implies (iw-2)\hat{y}(w) = \frac{1}{\sqrt{2\pi}}\frac{1}{2+iw}$$

$$\implies \hat{y}(w) = \frac{1}{\sqrt{2\pi}}\frac{1}{(2+iw)(-2+iw)} = -\frac{1}{\sqrt{2\pi}}\frac{1}{w^2+4}$$

$$\implies \hat{y}(w) = -\frac{1}{2\sqrt{2\pi}}\left(\frac{1}{w^2+4}\right)$$

Solution cont. . .

Taking inverse Fourier transform of both sides yields

$$y(t) = -\frac{1}{2\sqrt{2\pi}}\mathscr{F}^{-1}\left(\frac{1}{w^2+4}\right)$$

$$\implies y(t) = -\frac{1}{2\sqrt{2\pi}}\sqrt{\frac{\pi}{2}}e^{-2|t|} = \frac{e^{-2|t|}}{4}$$

Exercise 26:

Use the definition of the Fourier transform of an absolutely integrable function $f(x)$, to show that

$$\mathscr{F}(f'(x)) = iw\mathscr{F}(f(x))$$

Exercise 27:

If $\mathscr{F}(f(x)) = F(w)$ then show that

$$\mathscr{F}\{x^n f(x)\} = i^n F^{(n)}(w).$$

Exercise 28:

If $\mathscr{F}(f(x)) = F(w)$ and a is any real number then show that

$$\mathscr{F}(f(x-a)) = e^{-iwa}F(w).$$

Exercise 29:

Find the Fourier transform of $f(x) = xe^{-2x}, x > 0$.

Ans: $\hat{f}(w) = \frac{1}{\sqrt{2\pi}\ (2+iw)^2}$.

Exercise 30:

Show that $\mathscr{F}\left(\frac{1}{a^2+x^2}\right) = \frac{\sqrt{2\pi}}{2a}e^{-a|w|}$, where $a > 0$.

Solution

Consider the function $g(x) = e^{-a|x|}$. Then

$$\begin{aligned}
\hat{g}(w) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a|x|} e^{-iwx}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^{0} e^{ax} e^{-iwx}\, dx + \int_{0}^{\infty} e^{-ax} e^{-iwx}\, dx \right] \\
&= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{(a-iw)x}}{a-iw} \Big|_{-\infty}^{0} - \frac{e^{-(a+iw)x}}{a+iw} \Big|_{0}^{\infty} \right] \\
&= \frac{1}{\sqrt{2\pi}} \left[\frac{1}{a-iw} + \frac{1}{a+iw} \right] \\
&= \frac{1}{\sqrt{2\pi}} \left(\frac{2a}{a^2+w^2} \right)
\end{aligned}$$

Solution cont. . .

Now, taking the inverse Fourier transform, we have

$$\mathscr{F}^{-1}(\hat{g}(w)) = \mathscr{F}^{-1}\left(\frac{1}{\sqrt{2\pi}}\left(\frac{2a}{a^2+w^2}\right)\right) \implies e^{-a|x|} = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{2a}{a^2+w^2}e^{iwx}\,dw$$

$$\implies e^{-a|x|} = \frac{2a}{\sqrt{2\pi}}\left[\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{1}{a^2+w^2}e^{iwx}\,dw\right]$$

Changing the sign on x and multiplying by $\frac{\sqrt{2\pi}}{2a}$, we finally get

$$\frac{\sqrt{2\pi}}{2a}e^{-a|-x|} = \frac{\sqrt{2\pi}}{2a}e^{-a|x|} = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{e^{-iwx}}{a^2+w^2}\,dw = \hat{f}(x)$$

This gives

$$\hat{f}(w) = \frac{\sqrt{2\pi}}{2a}e^{-a|w|}$$

Thus,

$$\mathscr{F}\left(\frac{1}{a^2+x^2}\right) = \frac{\sqrt{2\pi}}{2a}e^{-a|w|}$$

Exercise 31:

Using the properties of the Fourier transform and a known Fourier transform computer the Fourier transforms of the following functions

(a) $f(x) = xe^{-x^2}$

(b) $f(x) = x^2 e^{-|x|}$

(c) $f(x) = \begin{cases} x & \text{if } |x| < 1 \\ 0 & \text{otherwise} \end{cases}$

(d) $f(x) = \begin{cases} xe^{-x} & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$

(e) $f(x) = \frac{x}{1+x^2}$

(f) $f(x) = \frac{x^2}{(1+x^2)^2}$

(g) $f(x) = (1-x^2)e^{-x^2}$

(h) $f(x) = (1-x^2)e^{-|x|}$

(i) $f(x) = xe^{-\frac{1}{2}(x-1)^2}$

(j) $f(x) = (1-x)e^{-|x-1|}$

Convolution Theorem

Convolution

The convolution of two functions f and g is denoted by $f * g$ and is defined by

$$(f * g)(x) = \int_{-\infty}^{\infty} f(t)g(x-t)\ dt = \int_{-\infty}^{\infty} g(t)f(x-t)\ dt.$$

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Remark

The following important properties of convolution can be proved easily

(i) $f * g = g * f$.

(ii) $f * [g * h] = [f * g] * h$.

(iii) $f * [g + h] = f * g + f * h$.

i.e., the convolution obeys the commutative, associative and distributive laws of algebra.

Convolution Theorem

Suppose that $f(x)$ and $g(x)$ are piecewise continuous, bounded, and absolutely integrable on the x-axis. Then

$$\mathscr{F}(f * g) = \sqrt{2\pi}\mathscr{F}(f)\mathscr{F}(g).$$

or Equivalently, if $\mathscr{F}(f) = F(w)$ and $\mathscr{F}(g) = G(w)$ then

$$\mathscr{F}^{-1}(F(w)G(w)) = \frac{1}{\sqrt{2\pi}}\left[\mathscr{F}^{-1}(F(w)) * \mathscr{F}^{-1}(G(w))\right]$$

Example

Suppose that f is an integrable and even ($f(-x) = f(x)$ for all x) and let $g(x) = \cos ax$. Show that, for all real numbers a:

$$f * g(x) = \sqrt{2\pi}\cos(ax)\hat{f}(a).$$

Solution

From the definition and the fact that $f * g = g * f$, we have

$$\begin{aligned} f * g(x) &= \int_{-\infty}^{\infty} f(t)\cos[a(x-t)]\ dt \\ &= \int_{-\infty}^{\infty} f(t)[\cos(ax)\cos(at) - \sin(ax)\sin(at)]\ dt \end{aligned}$$

Since f is an even function, the product $f(t)\sin(at)$ is odd; $\int_{-\infty}^{\infty} f(t)\sin(at)\ dt = 0$, and so

$$\begin{aligned} f * g(x) &= \cos(ax)\int_{-\infty}^{\infty} f(t)\cos(at)\ dt \\ &= \cos(ax)\int_{-\infty}^{\infty} f(t)[\cos(at) - i\sin(at)]\ dt \\ &= \cos(ax)\sqrt{2\pi}\left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(t)e^{-iat}\ dt\right) \\ &= \sqrt{2\pi}\cos(ax)\hat{f}(a) \end{aligned}$$

Example

For $n = 1, 2, \ldots$, let $g_n(x) = \begin{cases} \frac{n}{2} & \text{if } |x| < \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$. Suppose that f is continuous on $(-\infty, \infty)$ and F denotes an anti-derivative of f. Show that

$$f * g(x) = \frac{F\left(x + \frac{1}{n}\right) - F\left(x - \frac{1}{n}\right)}{2/n}.$$

Solution

From the definition, we have

$$f * g(x) = \int_{-\infty}^{\infty} f(x-t) g_n(t)\ dt$$
$$= \frac{n}{2} \int_{-\frac{1}{n}}^{\frac{1}{n}} f(x-t)\ dt$$

Making a change of variables: $x - t = u \implies dt = -du$, we get

$$f * g(x) = -\frac{n}{2} \int_{x+\frac{1}{n}}^{x-\frac{1}{n}} f(u)\ du$$
$$= -\frac{n}{2} \left[F\left(x - \frac{1}{n}\right) - F\left(x + \frac{1}{n}\right) \right]$$
$$= \frac{F\left(x + \frac{1}{n}\right) - F\left(x - \frac{1}{n}\right)}{2/n}$$

Exercise 32:

Let $f(x) = xe^{-\frac{x^2}{2}}$ and $g(x) = e^{-x^2}$.

(a) What are the Fourier transforms of f and g.

(b) What is the Fourier transform of $f * g$.

(c) Using part (b) find $f * g$.

Exercise 33:

Suppose that f is integrable and even and let $g(x) = \sin(ax)$. Show that, for all real numbers a, $f * g(x) = \sqrt{2\pi}\sin(ax)\hat{f}(a)$.

Exercise 34:

Suppose that f is integrable and odd and let $g(x) = \sin(ax)$. Show that, for all real numbers a, $f * g(x) = -i\sqrt{2\pi}\cos(ax)\hat{f}(a)$.

Applications of Fourier transforms in solving Partial differential equations

✻ **Laplace's Equation on the Half Plane:** We consider a steady state solution in two dimensions. In particular, we look for the steady state solution, $u(x, y)$, satisfying the two-dimensional Laplace equation on a semi-infinite slab with given boundary conditions as shown in Figure 6. The boundary value problem is given as

$$\begin{aligned} u_{xx} + u_{yy} = 0, \quad -\infty < x < \infty, \quad y > 0, \\ u(x, 0) = f(x), \quad -\infty < x < \infty \\ \lim_{y\to\infty} u(x, y) = 0, \quad \lim_{|x|\to\infty} u(x, y) = 0. \end{aligned}$$

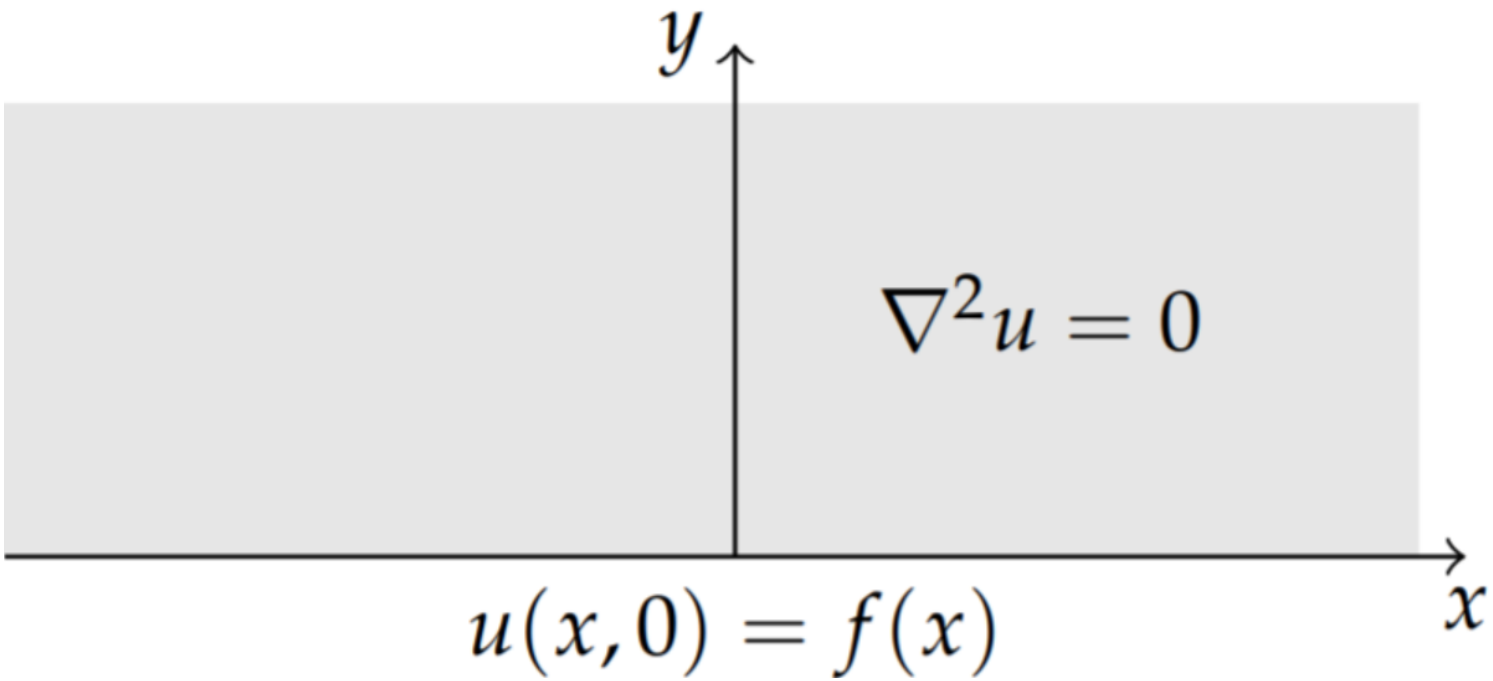


Figure 6: This is the domain for a semi-infinite slab with boundary value $u(x, 0) = f(x)$ and governed by Laplace's equation.

This problem can be solved using a Fourier transform of $u(x, y)$ with respect to x. The transform scheme for doing this is shown in Figure 7.

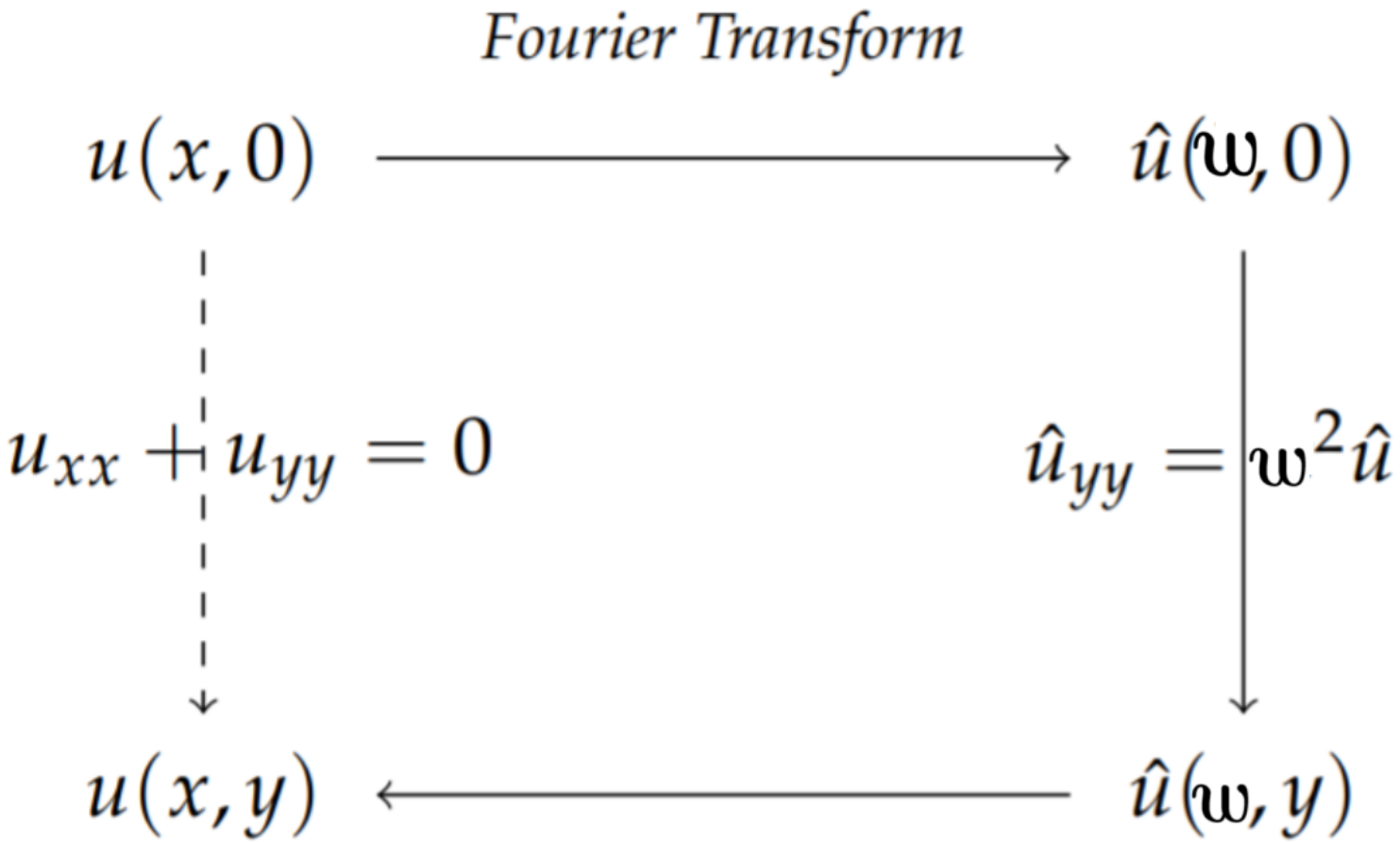


Figure 7: The transform scheme used to convert Laplace's equation to an ordinary differential equation which is easier to solve.

Example

The function $\phi = \phi(x, y)$ satisfies the Laplace equation in Cartesian coordinates

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0.$$

Use Fourier transforms to convert the above partial differential equation into an ordinary differential equation for $\hat{\phi}(w, y)$, where $\hat{\phi}(w, y)$ is the Fourier transform of $\phi(w, y)$ with respect to x.

Solution

We have

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0.$$

Taking the Fourier transform of this Pde yields

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\partial^2 \phi}{\partial x^2} e^{-iwx}\, dx + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\partial^2 \phi}{\partial y^2} e^{-iwx}\, dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} 0 e^{-iwx}\, dx.$$

Now

$$\mathscr{F}(f'(x)) = iw\mathscr{F}(f(x))$$
$$\mathscr{F}(f''(x)) = (iw)^2\mathscr{F}(f(x))$$

Substituting these values above, we have

Solution cont. . .

$$\frac{1}{\sqrt{2\pi}}(iw)^2 \int_{-\infty}^{\infty} \phi(x,y)e^{-iwx}\, dx + \frac{1}{\sqrt{2\pi}} \frac{\partial^2}{\partial y^2} \int_{-\infty}^{\infty} \phi(x,y)e^{-iwx}\, dx = 0$$

$$-w^2\hat{\phi}(w,y) + \frac{\partial^2}{\partial y^2}\left(\hat{\phi}(w,y)\right) = 0$$

w is a constant, so an ODE in y is

$$\frac{\partial^2}{\partial y^2}\left(\hat{\phi}(w,y)\right) - w^2\hat{\phi}(w,y) = 0$$

Example

The function $\phi = \phi(x, y)$ satisfies the Laplace equation in Cartesian coordinates

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0.$$

in part of the xy-plane for which $y \geq 0$. It is further given that

(a) $\lim_{y \to \infty} \phi(x, y) = 0, \quad \lim_{|x| \to \infty} \phi(x, y) = 0.$

(b) $\phi(x, 0) = \begin{cases} \frac{1}{2}, & |x| < 1 \\ 0, & |x| > 1 \end{cases}.$

Use Fourier transforms to show that

$$\phi(x, y) = \frac{1}{\pi} \int_0^\infty \frac{1}{w} e^{-wy} \sin w \cos wx \; dw,$$

and hence deduce the value of $\phi(\pm 1, 0)$.

Solution

We have

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0.$$

Taking the Fourier transform of this Pde yields

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\partial^2 \phi}{\partial x^2} e^{-iwx}\, dx + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\partial^2 \phi}{\partial y^2} e^{-iwx}\, dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} 0 e^{-iwx}\, dx.$$

$$\implies \frac{1}{\sqrt{2\pi}} (iw)^2 \int_{-\infty}^{\infty} \phi(x,y) e^{-iwx}\, dx + \frac{1}{\sqrt{2\pi}} \frac{\partial^2}{\partial y^2} \int_{-\infty}^{\infty} \phi(x,y) e^{-iwx}\, dx = 0$$

$$\implies -w^2 \hat{\phi}(w,y) + \frac{\partial^2}{\partial y^2} \left(\hat{\phi}(w,y) \right) = 0$$

$$\implies \frac{\partial^2}{\partial y^2} \left(\hat{\phi}(w,y) \right) - w^2 \hat{\phi}(w,y) = 0$$

$$\implies \hat{\phi}(w,y) = A(w) e^{-|w|y} + B(w) e^{|w|y}$$

as $\lim_{y\to\infty} \phi(x,y) = 0 \implies \lim_{y\to\infty} \hat{\phi}(w,y) = 0$. This gives $B(w) = 0$. Thus

$$\hat{\phi}(w,y) = A(w) e^{-|w|y}$$

Solution cont. . .

Next, we calculate the Fourier transform of $\phi(x,0) = g(x)$

$$\hat{\phi}(w,0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(x,0) e^{-iwx}\, dx = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} \frac{1}{2} e^{-iwx}\, dx$$

$$= \frac{1}{2\sqrt{2\pi}} \left[\frac{e^{-iwx}}{-iw} \right]_{-1}^{1} = \frac{1}{\sqrt{2\pi}} \frac{\sin w}{w}$$

This implies $\hat{\phi}(w,0) = A(w) = \frac{1}{\sqrt{2\pi}} \frac{\sin w}{w}$. This gives

$$\hat{\phi}(w,y) = \frac{1}{\sqrt{2\pi}} \frac{\sin w}{w} e^{-|w|y}.$$

Taking the inverse Fourier transform yields

$$\phi(x,y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \frac{\sin w}{w} e^{-|w|y} e^{iwx}\, dw$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \overbrace{\frac{\sin w}{w} e^{-|w|y}}^{\textit{even}} (\cos wx + i \sin wx)\, dw$$

Solution cont. . .

Therefore

$$\phi(x,y) = \frac{1}{\pi}\int_0^\infty \frac{\sin w}{w} e^{-wy}\cos wx \; dw.$$

Now,

$$\begin{aligned}\phi(\pm 1, 0) &= \frac{1}{\pi}\int_0^\infty \frac{\sin w}{w}\cos w \; dw \\ &= \frac{1}{2\pi}\int_0^\infty \frac{\sin 2w}{w} \; dw\end{aligned}$$

Substituting $2w = t \implies dw = \frac{dt}{2}$, we get

$$\begin{aligned}\phi(\pm 1, 0) &= \frac{1}{2\pi}\int_0^\infty \frac{\sin t}{t} \; dt \\ &= \frac{1}{2\pi} \times \frac{\pi}{2} = \frac{1}{4}\end{aligned}$$

Exercise 35:

The function $\phi = \phi(x, y)$ satisfies the Laplace equation in Cartesian coordinates

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0.$$

in the semi infinite region of the xy-plane for which $y \geq 0$. It is further given that

(a) $\lim_{y \to \infty} \phi(x, y) = 0, \quad \lim_{|x| \to \infty} \phi(x, y) = 0.$

(b) $\phi(x, 0) = f(x)$.

Use Fourier transforms to convert the above partial differential equation into an ordinary differential equation and hence show that

$$\phi(x, y) = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{f(x - u)}{u^2 + y^2} \, du.$$

✻ **Heat equation on an infinite interval:** We will consider the solution of the heat equation on an infinite interval using Fourier transforms. Consider the heat equation on the infinite line,

$$\begin{aligned} u_t &= \alpha u_{xx}, && -\infty < x < \infty, t > 0. \\ u(x,0) &= f(x), && -\infty < x < \infty. \end{aligned}$$

The basic scheme is outlined in Figure 8.

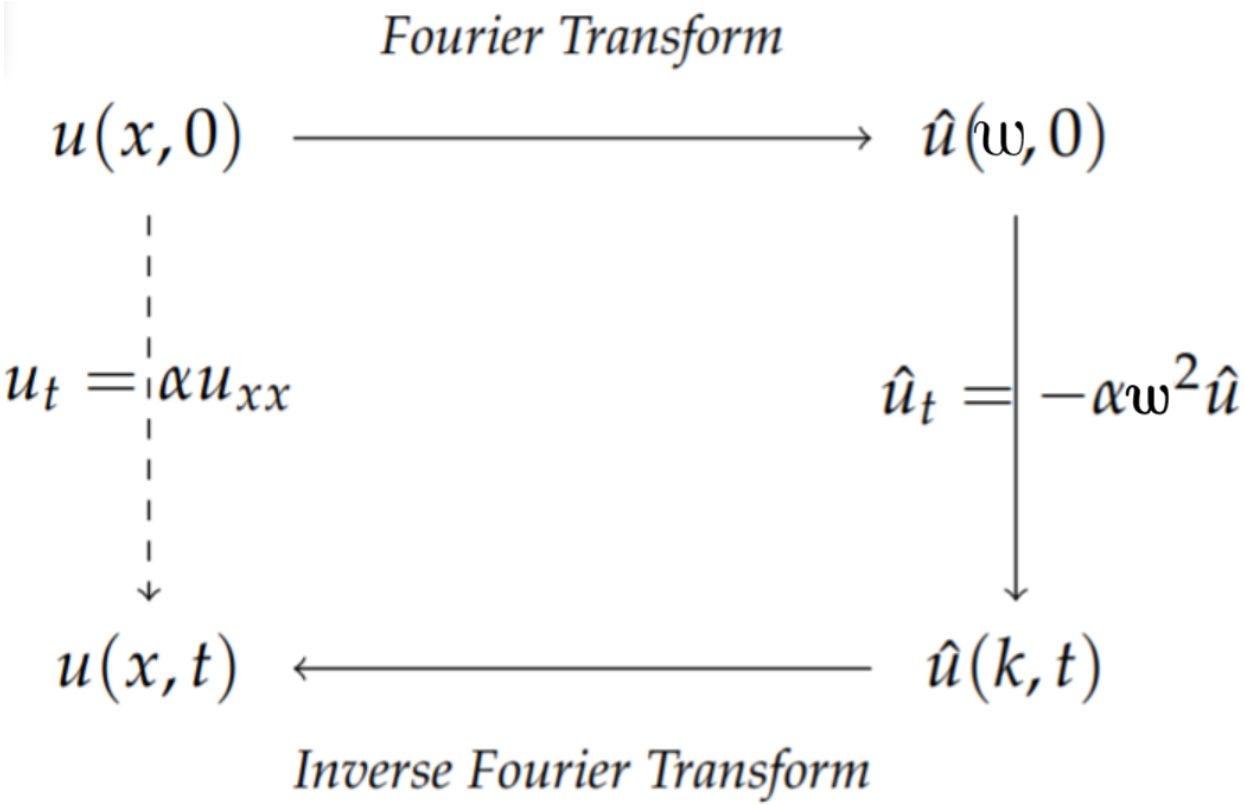


Figure 8: The transform scheme used to convert Heat equation to an ordinary differential equation which is easier to solve..

Example

The function $\theta = \theta(x, t)$ satisfies the heat equation in one spatial dimension,

$$\frac{\partial \theta}{\partial t} = \sigma^2 \frac{\partial^2 \theta}{\partial x^2}, \quad -\infty < x < \infty, \quad t \geq 0$$

where σ is a positive constant. Use Fourier transforms to convert the above partial differential into an ordinary differential equation.

Solution

$$\frac{\partial \theta}{\partial t} = \sigma^2 \frac{\partial^2 \theta}{\partial x^2}, \qquad \begin{matrix} -\infty < x < \infty, \\ t \geq 0 \end{matrix}$$ Subject to the initial condition $\theta(x, 0) = f(x)$.

Taking the Fourier transform of the P.D.E. in x, we get

$$\mathscr{F}\left[\frac{\partial \theta}{\partial t}\right] = \mathscr{F}\left[\sigma^2 \frac{\partial^2 \theta}{\partial x^2}\right]$$

$$\implies \frac{\partial}{\partial t}\mathscr{F}(\theta) = \sigma^2 \mathscr{F}\left[\frac{\partial^2 \theta}{\partial x^2}\right]$$

$$\implies \frac{\partial \hat{\theta}(w, t)}{\partial t} = \sigma^2 (iw)^2 \hat{\theta}(w, t)$$

w is a constant, so an ODE in t is

$$\frac{\partial \hat{\theta}(w, t)}{\partial t} = -\sigma^2 w^2 \hat{\theta}(w, t)$$

Example

The function $T = T(x,t)$ satisfies the heat equation in one spatial dimension,

$$\frac{\partial T}{\partial t} = \sigma \frac{\partial^2 T}{\partial x^2} \quad x \geq 0, \quad t \geq 0,$$

where σ is a positive constant. It is further given that

- $T(x,0) = f(x)$
- $T(0,t) = 0$
- $T(x,t) \to 0$ as $x \to \infty$.

Use Fourier transforms to convert the above partial differential equation into an ordinary differential equation and hence show that

$$T(x,t) = \frac{1}{\sqrt{4\pi\sigma t}} \int_{-\infty}^{\infty} f(u) \exp\left(-\frac{(x-u)^2}{4t\sigma}\right) \, du.$$

You may assume that $\mathscr{F}\left(e^{-ax^2}\right) = \frac{1}{\sqrt{2a}} e^{-\frac{w^2}{4a}}$.

Solution

$$\frac{\partial T}{\partial t} = \sigma \frac{\partial^2 T}{\partial x^2}, \qquad x \geq 0, \quad t \geq 0.$$

Taking the Fourier transform of the P.D.E. in x, we get

$$\mathscr{F}\left[\frac{\partial T}{\partial t}\right] = \mathscr{F}\left[\sigma \frac{\partial^2 T}{\partial x^2}\right]$$

$$\implies \frac{\partial}{\partial t}\left(\mathscr{F}\left(T\right)\right) = \sigma \mathscr{F}\left[\frac{\partial^2 T}{\partial x^2}\right]$$

$$\implies \frac{\partial \hat{T}(w,t)}{\partial t} = \sigma (iw)^2 \hat{T}(w,t)$$

$$\implies \frac{\partial \hat{T}(w,t)}{\partial t} = -\sigma w^2 \hat{T}(w,t)$$

Thus, we have an ODE of first order in $\hat{T}(w,t)$, w is treated as a constant. Separating the variables, we get

$$\frac{\partial \hat{T}(w,t)}{\hat{T}(w,t)} = -\sigma w^2 \partial t \implies \hat{T}(w,t) = A(w) e^{-\sigma w^2 t}$$

Solution cont. . .

Now the initial condition

$$T(x,0) = f(x) \implies \hat{T}(w,0) = \hat{f}(w).$$

gives $A(w) = \hat{f}(w)$. Thus, we have

$$\hat{T}(w,t) = \hat{f}(w)e^{-\sigma w^2 t}$$

Taking the inverse Fourier transform yields

$$\begin{aligned}
\mathscr{F}^{-1}[\hat{T}(w,t)] &= \mathscr{F}^{-1}[\hat{f}(w)e^{-\sigma w^2 t}] \\
\implies T(x,t) &= \frac{1}{\sqrt{2\pi}}\left(\mathscr{F}^{-1}[\hat{f}(w)] * \mathscr{F}^{-1}[e^{-\sigma w^2 t}]\right) \\
\implies T(x,t) &= \frac{1}{\sqrt{2\pi}}\left(f(x) * \mathscr{F}^{-1}[e^{-\sigma w^2 t}]\right) \qquad (7)
\end{aligned}$$

Solution cont. . .

Given that

$$\mathscr{F}\left(e^{-ax^2}\right) = \frac{1}{\sqrt{2a}} e^{-\frac{w^2}{4a}} \implies \mathscr{F}^{-1}[e^{-\frac{w^2}{4a}}] = \sqrt{2a}\left(e^{-ax^2}\right)$$

This gives

$$\mathscr{F}^{-1}[e^{-\sigma w^2 t}] = \frac{1}{\sqrt{2\sigma t}} e^{-\frac{x^2}{4\sigma t}}$$

Substituting the value of $\mathscr{F}^{-1}[e^{-\sigma w^2 t}]$ in (7), we get

$$T(x,t) = \frac{1}{\sqrt{2\pi}}\left(f(x) * \frac{1}{\sqrt{2\sigma t}} e^{-\frac{x^2}{4\sigma t}}\right) = \frac{1}{\sqrt{4\pi\sigma t}}\left(f(x) * e^{-\frac{x^2}{4\sigma t}}\right)$$

$$\implies T(x,t) = \frac{1}{\sqrt{4\pi\sigma t}} \int_{-\infty}^{\infty} f(u) \exp\left(-\frac{(x-u)^2}{4t\sigma}\right) du.$$

Thus

$$\boxed{T(x,t) = \frac{1}{\sqrt{4\pi\sigma t}} \int_{-\infty}^{\infty} f(u) \exp\left(-\frac{(x-u)^2}{4t\sigma}\right) du}$$

Exercise 36:

The function $\theta = \theta(x, t)$ satisfies the heat equation in one spatial dimension,

$$\frac{\partial \theta}{\partial t} = \sigma^2 \frac{\partial^2 \theta}{\partial x^2}, \quad -\infty < x < \infty, \quad t \geq 0$$

where σ is a positive constant. Given that $\theta(x, 0) = f(x)$, use Fourier transforms to convert the above partial differential into an ordinary differential equation and hence show that

$$\theta(x, t) = \frac{1}{2\sigma\sqrt{\pi t}} \int_{-\infty}^{\infty} f(x - u) \exp\left(-\frac{u^2}{4t\sigma^2}\right) \, du.$$

Exercise 37:

The temperature $u(x,t)$ at any point of an infinite bar satisfies the equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad -\infty < x < \infty, \quad t \geq 0$$

and the initial temperature along the length of the bar is given by

$$u(x,0) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases}.$$

Ans: $u(x,t) = \frac{2}{\pi} \int_0^\infty e^{-w^2 t} \frac{\sin w \cos wx}{w} \, dw$

Exercise 38:

An infinite string is initially at rest and its initial displacement is given by $f(x)$, $-\infty < x < \infty$. Determine the displacement $y(x,t)$ of the string, where the equation of the vibrating string is given by

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}.$$

Solution

The equation of the vibrating string is given by

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2},$$

with the initial conditions

(i) $\left.\frac{\partial y}{\partial t}\right|_{t=0} = 0$ (ii) $y(x,0) = f(x)$

Taking the Fourier transforms on both sides with respect to x, we get

$$\mathscr{F}\left(\frac{\partial^2 y}{\partial t^2}\right) = c^2 \mathscr{F}\left(\frac{\partial^2 y}{\partial x^2}\right)$$

$$\implies \frac{\partial^2 \hat{y}(w,t)}{\partial t^2} = c^2 (iw)^2 \hat{y}(w,t) \quad \text{where} \quad \mathscr{F}(y(x,t)) = \hat{y}(w,t)$$

$$\implies \frac{d^2 \hat{y}(w,t)}{dt^2} + c^2 w^2 \hat{y}(w,t) = 0,$$

Solution cont. . .

which is ordinary differential equation of second order. Solution to this is given by

$$\hat{y}(w,t) = A\cos(cwt) + B\sin(cwt).$$

Putting $t = 0$ in above expression yields

$$\hat{y}(w,0) = A.$$

Given that $y(x,0) = f(x) \implies \hat{y}(w,0) = \hat{f}(w) = A$. Thus

$$\hat{y}(w,t) = \hat{f}(w)\cos(cwt) + B\sin(cwt)$$

$$\implies \frac{\partial \hat{y}(w,t)}{\partial t} = -cw\hat{f}(w)\sin(cwt) + Bcw\cos(cwt) \implies \left.\frac{\partial \hat{y}(w,t)}{\partial t}\right|_{t=0} = cwB$$

Also, given that

$$\left.\frac{\partial y(x,t)}{\partial t}\right|_{t=0} = 0 \implies \left.\frac{\partial \hat{y}(w,t)}{\partial t}\right|_{t=0} = 0$$

$$\implies cwB = 0 \implies B = 0$$

Therefore

$$\hat{y}(w,t) = \hat{f}(w)\cos(cwt)$$

Taking inverse Fourier transform, we have

$$y(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(w)\cos(cwt)e^{iwx}\ dw$$